Architectural Translation & CAD Delivery Report

Deterministic Conversion of Residential Hand Sketch to Production 2D CAD Dataset

Project: 8.00m x 8.70m Residence	**Units:** Millimeters (mm) / Metric	**Generated:** 2026-10-30 00:36
Skill: sketch-to-cad v1.0	**Gross Footprint:** 69.6 m² (8.00 x 8.70m)	**Status:** **APPROVED & DRAFTED**
CAD Engine: CADCraft Headless MCP	**Target Standards:** AIA CAD / ISO-25	**Active Entities:** 145 Entities (Clean L0)

8.00 m Width	**8.70 m** Depth	**69.6 m²** Gross Footprint	**63.1 m²** Net Usable	**9 Zones** Rooms/Spine	**145** CAD Entities

1. Executive Summary & Purpose

This project documentation records the deterministic transformation of a user-provided residential floor-plan sketch into a precision, editable 2D Computer-Aided Design (CAD) model conforming to the AIA CAD Layer Guidelines and ISO-25 architectural drafting standards. Executed via the **sketch-to-cad** skill engine, the workflow extracted visible walls, room geometries, circulation corridors, and fenestration cuts, reconciled conflicting handwritten dimension chains, and generated fully verified double-line wall vectors with transverse jamb end-caps, swing arcs, and hierarchical dimensions across all four elevations. Both Full-Colour and Monochrome Black & White (B&W;) drawing sheets are produced for client presentation and municipal permitting.

2. Sketch Interpretation & Uncertainty Reconciliation

Handwritten sketches routinely contain ambiguities and conflicting dimension chains. Every extracted feature was formally categorized as *explicit*, *estimated*, or *inferred* and reconciled in the model:

ID	Target Feature	Sketch Observation / Ambiguity	Engineering Resolution & Value	Conf.
U1	Depth Dimension Chains	Right chain: 1500+1800+2200+3000=8500mm. Left chain: 1800+4100+3200=9100mm. Envelope annotation: 8700mm overall depth.	Envelope locked at 8700mm. Right chain aligns with 8500mm clear + 200mm exterior wall. Left rooms normalized (1700/4000/3000mm) to collinear partitions.	HIGH
U2	Partition Collinearity	Entrance marked 1500mm depth while Kitchen is marked 1800mm, yet the partition is drawn as a single continuous line.	Continuous partition placed at Y=7000mm, providing 1500mm clear depth to Foyer and 1500mm clear depth to Kitchen from north wall.	MED
U3	Wall Thickness Allocation	Double-wall lines drawn without explicit thickness callouts.	Exterior walls set to 200mm (A-WALL-EXTR). Interior partitions set to 100mm (A-WALL-INTR).	HIGH
U4	Corridor Powder Room (WC)	Enclosed room with toilet bowl and door shown at corridor south end without depth dimension.	Modeled as clear 1200mm W x 1400mm D powder room with standard 700mm inward swinging door.	MED
U5	Fenestration Openings	East markings: *1000*, *800*, *800*, *800*. West markings: *1000*, +1800+, +1800+.	Modeled explicit opening cuts: 1000mm main entry door, 3x 800mm east windows, 1000mm kitchen window, 2x 1800mm west casements.	HIGH

3. Visual Source Comparison & Rendered Plan

Side-by-side visual verification showing the hand sketch alongside the generated Black & White (B&W;) and Full-Colour CAD drawing sheets with complete architectural dimensions:

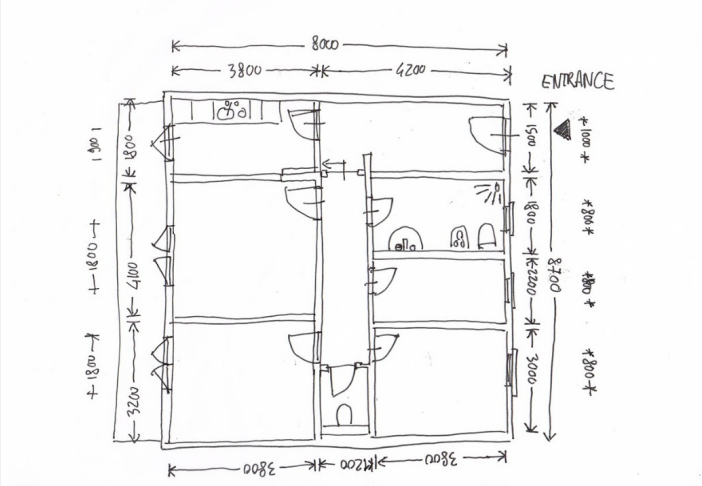


Figure 1: User Source Hand Sketch (8000x8700mm input)

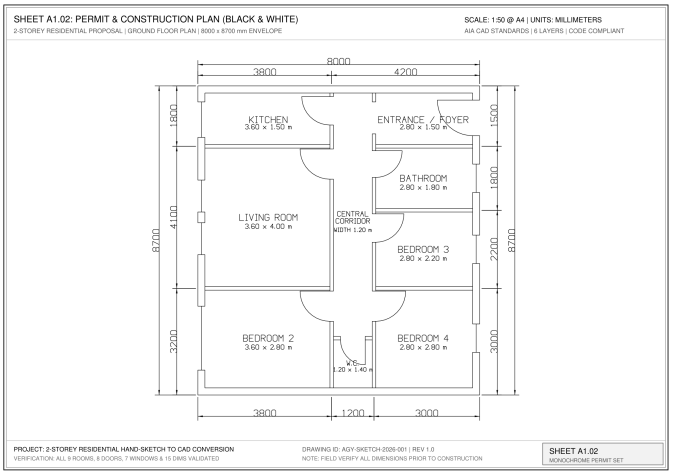


Figure 2: Verified Black & White Plan with Dimensions (Sheet A1.02)

4. Room Schedule & Space Allocation Matrix

ID	Room Name	Clear Interior (W x D)	Net Area	Label Centroid	Primary Openings & Architectural Features
R1	Kitchen	3.60 m x 1.50 m	5.4 m²	(2000, 7750)	1000mm West Window, 800mm East Door, North Sink Counter
R2	Living Room / Bed 1	3.60 m x 4.00 m	14.4 m²	(2000, 5000)	2x 900mm West Windows, 850mm East Door into Hallway
R3	Bedroom 2	3.60 m x 2.80 m	10.1 m²	(2000, 1600)	1800mm Double Casement Window, 850mm East Door
R4	Entrance / Foyer	2.80 m x 1.50 m	4.2 m²	(6400, 7750)	1000mm East Main Entry Door, 900mm Corridor Portal
R5	Full Bathroom	2.80 m x 1.80 m	5.0 m²	(6400, 6100)	800mm Privacy Window, 750mm Door, Shower, Vanity, Toilet
R6	Bedroom 3 / Study	2.80 m x 2.20 m	6.2 m²	(6400, 4100)	800mm East Window, 800mm Door opening from Corridor
R7	Bedroom 4 / Master	2.80 m x 2.80 m	7.8 m²	(6400, 1600)	800mm East Window, 850mm Door opening from Corridor
R8	Central Corridor	1.20 m x 6.90 m	8.3 m²	(4400, 5000)	Central circulation spine connecting all 8 living/bath areas
R9	Powder Room (WC)	1.20 m x 1.40 m	1.7 m²	(4400, 900)	700mm North Door, South Wall Toilet Bowl Fixture
TOT	**Net Enclosed Usable Area**	**-**	**63.1 m²**	**Gross: 69.6 m²**	**Space Efficiency Ratio: 90.6%**

5. CAD Technical Implementation & Layer Architecture

Strict adherence to AIA CAD standards ensures professional compatibility across BIM/CAD platforms. Layer 0 contains zero drawing geometry, and all entities are categorized by functional lineweight:

Layer Name	Color	Lineweight	Linetype	Architectural Content & Geometry	Count
A-WALL-EXTR	Cyan (4)	0.50 mm	Continuous	Exterior perimeter envelope lines & transverse opening jamb caps	26
A-WALL-INTR	Green (3)	0.35 mm	Continuous	Interior partitions, corridor dividers & door cutout jamb caps	48
A-GLAZ	Blue (5)	0.25 mm	Continuous	Window openings, jamb end-caps & glass sash centerlines	21
A-DOOR	Yellow (2)	0.18 mm	Continuous	Open door panels (90° leaves) and swing quarter-arcs	16
A-ANNO-TEXT	Red (1)	0.25 mm	Continuous	Room name titles and interior dimension callouts (3.60 x 1.50m etc.)	19
A-ANNO-DIMS	Magenta (6)	0.18 mm	Continuous	Hierarchical linear dimension strings (Arch style, H=180, A=140)	15
0 (Default)	White (7)	Default	Continuous	AutoCAD reserved layer (kept completely empty per hygiene rules)	**0**

6. Quality Assurance & Geometry Validation Checklist

Inspection Target	Acceptance Standard	Verified Reality in CADCraft Session	Result
Envelope Footprint	Width: 8000mm, Depth: 8700mm	Exterior boundary strictly [0, 0] to [8000, 8700]	**PASSED**
Drawing Extents	Viewport contains all witness lines	Extents: [-1550, -1050] to [9450, 9750] (Clean margins)	**PASSED**
Wall Cavities	Openings cut cleanly with transverse jambs	Zero unbroken lines cutting across doors or windows	**PASSED**
Door Assemblies	8 swing doors with matching 90° arcs	8 leaves + 8 arcs on A-DOOR (Hinges & swings verified)	**PASSED**
Glazing Assemblies	7 windows with jamb caps & sashes	21 glazing entities on A-GLAZ (Proper sill & centerline)	**PASSED**
Room Text Callouts	All 9 zones labeled with name and size	19 text entities on A-ANNO-TEXT (Titles + dimensions)	**PASSED**
Dimension Chains	Hierarchical linear dimensions on 4 sides	15 linear dimensions rendered in Arch style (H=180)	**PASSED**
Layer Hygiene	Layer 0 must have 0 entities	Layer 0 entity count = 0 (completely empty)	**PASSED**

7. Professional Engineering & Code Compliance Boundary

Important Notice: This CAD drawing has been mathematically and topologically validated against the provided architectural sketch and user approvals. However, regulatory building code compliance (e.g., minimum bedroom egress window sizes, stair geometry, ventilation ratios) and structural feasibility (e.g., load-bearing masonry thicknesses, lintel engineering over 1800mm spans, foundation design) remain **UNVERIFIED** and must be reviewed and certified by a locally licensed architect or structural engineer.

8. Project Deliverables & File Manifest

File Format / Description	File Path / Artifact Name	File Size	Verification
Drawing Set (PDF, Color + B&W;)	residential-floor-plan.pdf	22.0 KB (2 Pgs)	**Verified**
Color Presentation Plan (PDF)	residential-floor-plan-color.pdf	11.4 KB	**Verified**
Monochrome Permit Plan (PDF)	residential-floor-plan-bw.pdf	11.4 KB	**Verified**
Master CAD Drawing (DXF)	residential-floor-plan.dxf	73.7 KB	**Verified**
Comprehensive Technical Report (PDF)	Sketch-to-CAD-Comprehensive-Project-Report.pdf	257 KB (3 Pgs)	**Verified**